# The Math Behind Spatial Pattern Formation

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**Abstract**

Spatial patterns are present throughout many areas of nature such as in animal patterns and vegetation growth. Turing patterns are one way of mathematically modeling spatial pattern formation, and the reaction-diffusion principles which define the model have been shown to be the likely biological cause of pattern formation in animals. Due to their direct tie to the natural phenomenon they represent, Turing patterns are a significant area of interest in mathematical biology. The pattern formation arises from complex interactions between two or more substances diffusing into each other. With proper analysis, the dynamics can be made to generate an infinite number of varying patterns. This work studies the Schnakenberg kinetics, known to generate spot patterns, and performs a linear stability analysis to determine the necessary conditions for generating Turing patterns. Parameters are chosen based on this analysis to simulate spot patterns similar to those seen on a cheetah.

**Keywords:** Turing pattern, reaction-diffusion, linear stability

## 1. Introduction

Diffusion is generally considered spatially stabilizing, since a substance will start with a certain concentration and spread out to reach an equilibrium. For example, imagine a diffusing mound of sand. As time goes on, we expect the mound to will flatten out. Now imagine instead that the mound of sand spreads unevenly and forms into multiple smaller mounds spread out over the domain. This is what is known as Turing instability. In 1952, Alan Turing proposed that under the right conditions, diffusion of two or more substances can actually be destabilizing and lead to the formation of spatial patterns [1]. When a system exhibits Turing instability, the steady state is linearly stable to small perturbations in the absence of diffusion and unstable to small spatial perturbations in the presence of diffusion [2]. The first requirement means that without diffusion, the substances will spread out to a homogeneous state and stay there. The second requirement means that when diffusion is added back in, the substances will start diffusing into each other and patterns can arise from the push and pull of their interactions. Linear stability without diffusion guarantees that it is the diffusion terms which create the patterns.

Turing originally proposed the idea of reaction-diffusion equations to describe chemical Morphogenesis, a process by which structures develop shape. The principle has since been applied to pattern formation across scientific fields, with prominent applications in vegetation growth, human biology, and animal spot pattern formation. The beauty of Turing pattern theory comes from its inextricable link to the natural phenomenon that it describes. While initially written to model chemical interactions, the principle has since been used to theorize about the underlying natural causes for biological processes. For example, morphogenesis of preexisting concentrations of cells in animals has been shown to be the driving cause of animal spot formation [3][4].

In this work, we derive the necessary conditions for Turing instability in a reaction-diffusion system and apply this analysis to generating a pattern of spots reminiscent of a cheetah coat. The main text followed is *Mathematical Biology* chapter 14 by J.D. Murray [2]. For applying the theory to generating a spot pattern using the well-known Schnakenberg kinetics, the main text followed is *Pattern Formation Analysis in the Schnakenberg Model* by Casper H. L. Beentjes [5].

## 2. Reaction-Diffusion Kinetics

### 2.1. Basic Model

A reaction-diffusion system representing the concentration of two substances over time and space takes the form of the following partial differential equations (PDEs),

$$\begin{aligned} A_T(X,Y,T) &= F(A,B) + D_A\nabla^2 A, \\ B_T(X,Y,T) &= G(A,B) + D_B\nabla^2 B. \end{aligned} \tag{1}$$

Here, $X$ and $Y$ are spatial variables and $T$ represents time. The subscript $T$ represents the partial derivative operator $\frac{\partial}{\partial T}$ and $\nabla^2$ represents the Laplace operator in two dimensions $(X, Y)$, also written $\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}$. These equations

describe the changes in concentration of substances $A$ and $B$ over time and space. $F$ and $G$ are functions which dictate the reaction kinetics. In simpler terms, they describe relationship between the two substances. $D_A$, $D_B > 0$ are the diffusion coefficients which dictate the rates of diffusion. Any equation in the form of Eq. (1) can be written in the non-dimensional form

$$\begin{aligned} u_t(x,y,t) &= \gamma f(u,v) + \nabla^2 u, \\ v_t(x,y,t) &= \gamma g(u,v) + d\nabla^2 u \end{aligned} \tag{2}$$

[2]. The non-dimensional form is, intuitively, a system of reaction-diffusion equations similar to the original system.

For simplicity, in this paper we work on a square domain. We must also prescribe boundary conditions on the domain. The boundary conditions are important to the dynamics and impact the pattern formation [2][6]. In this study, we apply zero-flux boundary conditions to the system. These are Neumann conditions which enforce that rate of change of the substance with respect to the spatial variables is zero. Such boundary conditions mean that we are not dictating the concentration of the substances at the boundaries which would impact the patterns generated [2]. Instead, we are imposing the condition that at the boundary, nothing external is interfering with the system and the substances are not leaving the domain. We also need to prescribe an initial condition. The choice of condition is discussed in Sections 5.1 and 6.3.

### 2.2. Physical Implications of Reaction Kinetics

Many functions can be chosen to describe the reaction kinetics of the system. Most models which aim to generate Turing patterns choose activator-inhibitor kinetics, although other carefully chosen kinetics can also generate Turing patterns [7]. In models of the traditional activator-inhibitor type, one substance, known as the activator, acts as a catalyst for its own production (autocatalytic) but also depends on the other substance for continued production. The other substance, or inhibitor, feeds the activator, but also slows down the activator's production if its own concentration gets too small.

Another factor to take into consideration when defining reaction kinetics is the desired pattern type. Different nonlinear terms have been found to produce different types of patterns [8].

### 2.3. Schnakenberg Kinetics

The reaction kinetics studied in this paper describe simple activator-inhibitor behavior and are known as the Schnakenberg Kinetics. These kinetics include a quadratic term which is known to favor spot patterns [8], making them suitable for modeling patterns such as cheetah spots. The reaction terms are defined by

$$\begin{aligned} F(A,B) &= k_2 - k_2 A + k_3 A^2 B, \\ G(A,B) &= k_4 - k_3 A^2 B, \end{aligned}$$

where the $k$'s are positive rate constants. Here, substance $A$ is considered to be the activator and substance $B$ is considered the inhibitor. This is because substance $A$ is a catalyst for itself, as represented by the term $k_3 A^2 B$ appearing in the equation for $F$. The presence of substance $B$ in the same term also means that by decreasing too much in concentration, substance $B$ inhibits the growth of substance $A$. This joint decrease then promotes the growth of both substances. These cyclical dynamics are shown in Figure 1.

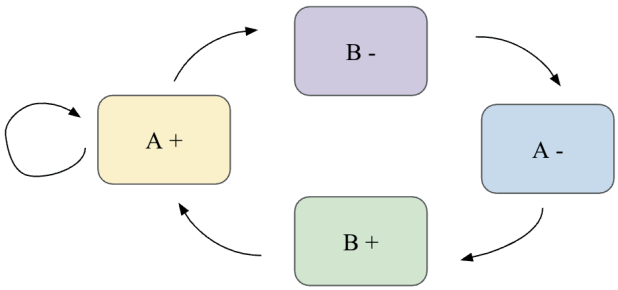



Figure 1: Schankenberg activator-inhibitor kinetics. Substances $A$ and $B$ have a cyclical relationship. Substance $A$ is an activator and promotes its own growth. Substance $B$ is an inhibitor and does not allow substance $A$ to grow unchecked.

In nondimensional form, the reaction-diffusion system can be written as

$$\begin{aligned} u_t(x,y,t) &= \gamma(a - u + u^2 v) + \nabla^2 u, \\ v_t(x,y,t) &= \gamma(b - u^2 v) + d\nabla^2 u. \end{aligned} \tag{3}$$

The parameters under the nondimensionalization with length scale $L$ are

$$u = A\left(\frac{k_3}{k_2}\right)^{\frac{1}{2}},\ v = B\left(\frac{k_3}{k_2}\right)^{\frac{1}{2}},\ t = \frac{D_A T}{L^2},\ x = \frac{X}{L},$$

$$d = \frac{D_B}{D_A},\ a = \frac{k_1}{k_2}\left(\frac{k_3}{k_2}\right)^{\frac{1}{2}},\ b = \frac{k_4}{k_2}\left(\frac{k_3}{k_2}\right)^{\frac{1}{2}},\ \gamma = \frac{L^2 k_2}{D_A}.$$

## 3. Conditions for Turing Instability

Pattern formation occurs under specific conditions and not all systems will generate patterns [2]. The functions $f$ and $g$ which define the reaction kinetics can have various forms and parameters. For example, in the case of the

nondimesional Schnakenberg kinetics, only some choices of parameters $a$, $b$, and $d$ will define a system capable of producing patterns. In order to find these parameters, we can perform a linear stability analysis. Recall that Turing patterns are capable of forming when the system is linearly stable in the absence of diffusion and unstable to spatial perturbations in the presence of diffusion.

While such a linear analysis is effective in informing parameter choices, the non-linear dynamics of the reaction-diffusion system will still play a significant role in the formation of patterns. We can consider the linear stability analysis to inform us of the baseline requirements for the system, while working under the knowledge that the non-linear terms will result in dynamics which are more difficult to predict. We begin by conducting the linear stability analysis on a generic function in the form of Eq. (2).

### 3.1. Stability in the Absence of Diffusion

We look first to enforce linear stability in the absence of diffusion. Without the diffusion terms, the system defined in Eq. (2) takes the form

$$u_t = \gamma f(u,v),\ v_t = \gamma g(u,v).$$

Since $f$ and $g$ are each twice differentiable with respect to $u$ and $v$, we know that the system is locally linear at a steady state. This means that if the system is initially close to the steady state, denoted $(u_*, v_*)$, then we can describe a linear system such that near said state, the behavior will resemble that of the non-linear system. To find the steady state, we set $u_t = \gamma f(u,v) = 0$ and $v_t = \gamma g(u,v) = 0$.

Without the diffusion term, we can linearize about the steady state by setting $\mathbf{w} = [u - u_*, v - v_*]^T$. Then the system can be written in matrix-vector form as

$$\mathbf{w}_t = \gamma J_{(u_*,v_*)}\mathbf{w},$$

where $J_{(u_*,v_*)}$ is the Jacobian matrix evaluated at the steady state. The stability of a linear system can be determined by the eigenvalues $\lambda$ of the matrix $\gamma J$. A linearly stable steady state will have $Re(\lambda) < 0$. Since the trace of the matrix is equal to the sum of the eigenvalues and the determinant is equal to the product of the eigenvalues, we can enforce stability by requiring that $\text{tr}(\gamma J) < 0$ and $\det(\gamma J) > 0$. The trace is given by $\text{tr}(\gamma J) = \gamma(f_u + g_v)$. The determinant is given by

$$\det(\gamma J) = \begin{vmatrix} \gamma f_u & \gamma f_v \\ \gamma g_u & \gamma g_v \end{vmatrix} = \gamma^2(f_u g_v - f_v g_u).$$

Since $\gamma > 0$, we need only require $\text{tr}(J) < 0$ and $\det(J) > 0$. Therefore, it must be the case that

$$f_u + g_v < 0, \tag{C.1}$$

$$f_u g_v - f_v g_u > 0. \tag{C.2}$$

Under these conditions, the steady state will be an asymptotically stable node, an asymptotically stable spiral, or a proper or improper node.

Condition (C.1) is especially interesting in terms of physical meaning. The concentrations of substances $u$ and $v$ over time are directly tied to the behavior of the functions $f$ and $g$. Physically, $f_u$ describes how the reaction kinetics change in response to changes in the concentration of substance $u$, and $g_v$ describes changes in function $g$ with regards to the concentration of substance $v$. This means that if $f_u$ is positive, then increase in substance $u$ causes more increase in substance $u$. In other words, $u$ is autocatalytic. The sum $f_u + g_v$ must be negative, so if $f_u$ is positive, $g_v$ must be negative. This means substance $v$ cannot be autocatalytic. Therefore, only one substance in can be autocatalytic if patterns are to be formed. This shows us a reason why activator-inhibitor kinetics are the typical model used to generate Turing patterns.

### 3.2. Diffusion-Driven Instability

Now that we have enforced stability in the absence of diffusion, we turn to inducing instability in the presence of diffusion. We add diffusion back in to the system and once again linearize about the steady state $(u_*, v_*)$. The locally linear system for Eq. (2) is of the form

$$\mathbf{w_t} = \gamma J_{(u*,v_*)}\mathbf{w} + D\nabla^2\mathbf{w},\ \text{for } D = \begin{bmatrix} 1 & 0 \\ 0 & d \end{bmatrix}.$$

This is a significantly more complicated system than the one discussed in Section 3.1, as it is a system of PDEs which involves derivatives in both time and space. As before, we are not looking for a solution to the system, but instead for conditions on the eigenvalues which, when enforced, will cause pattern formation. Since we are interested in inducing spatial instability, we start by separating the equations in time and space.

Assume a separable solution $\mathbf{w}(t,\mathbf{x}) = T(t)\mathbf{X}(\mathbf{x})$ where $\mathbf{x} = [x,y]^T$. Then the system takes the form

$$T'(t)\mathbf{X}(\mathbf{x}) = \gamma J T(t)\mathbf{X}(\mathbf{x}) + D\nabla^2 T(t)\mathbf{X}(\mathbf{x}).$$

Separating by dividing both sides by $T(t)\mathbf{X}(\mathbf{x})$ we get,

$$\frac{T'(t)}{T(t)} = \frac{\gamma J\mathbf{X}(\mathbf{x}) + D\nabla^2\mathbf{X}(\mathbf{x})}{\mathbf{X}(\mathbf{x})} = \lambda,$$

keeping in mind that we have Neumann boundary conditions in $\mathbf{X}(\mathbf{x})$ and an initial condition in $T(t)$. The result of the separation of variables is an ODE in $t$,

$$T'(t) = \lambda T(t), \tag{4}$$

with solution $T(t) = e^{\lambda t}$, and another system of PDEs in $\mathbf{x}$,

$$\gamma J\mathbf{X}(\mathbf{x}) + D\nabla^2\mathbf{X}(\mathbf{x}) = \lambda\mathbf{X}(\mathbf{x}), \tag{5}$$

the spatial problem we were looking for.

Eq. (5) is a form of what is known as the Strum-Liouville eigenvalue problem. It is known that problems of this form can be expressed as an absolutely and uniformly convergent series in terms of the eigenfunctions [9]. Therefore, we can write

$$\mathbf{X}(\mathbf{x}) = \sum_{k}^{\infty} \mathbf{a_k} \mathbf{w_k}(\mathbf{x}),$$

where $\mathbf{w_k}(\mathbf{x})$ are the eigenfunctions and thus satisfy the property

$$\nabla^2 \mathbf{w_k}(\mathbf{x}) = -Dk^2 \mathbf{w_k}(\mathbf{x}). \tag{6}$$

Then we can rewrite Equation (5) using the relation given in Equation (6) as

$$\lambda_k \mathbf{w_k}(\mathbf{x}) = (-Dk^2 + \gamma J)\mathbf{w_k}(\mathbf{x}),$$

where $\lambda_k$ is a function of the new variable $k$ which is considered the wavenumber since it relates to the frequency of the pattern. This is an eigenvalue problem in the classical sense, where $\lambda_k$ is the eigenvalue and $\mathbf{w_k}(\mathbf{x})$ is the eigenvector. Thus $\lambda_k$ is the solution to $\det(-Dk^2 + \gamma J - \lambda_k I) = 0$.

Now that we have an expression for the eigenvalue $\lambda_k$, we are at the stage where we can determine conditions on the reaction kinetics which will cause the desired stability/instability.

First, note that when $k = 0$, we are in the linearly stable steady state caused by the absence of diffusion. The characteristic polynomial is then determined by $\det(\gamma J - \lambda_0 I) = 0$ and the trace and determinant will need to satisfy the stability conditions derived in Section 3.1.

When $k \neq 0$, we want to impose instability. This will occur when $Re(\lambda_k) > 0$. To satisfy this, we require either $\mathrm{tr}(-Dk^2 + \gamma J) > 0$ or $\det(-Dk^2 + \gamma J) < 0$. Looking first at the trace, we need

$$\mathrm{tr}(-Dk^2 + \gamma J) = \gamma(f_u + g_v) - k^2(d+1) > 0.$$

However, in Section 3.1, we required that $\gamma(f_u + g_v) < 0$ (Condition (**C.1**)) in order to guarantee linear stability, and since $k^2$ and $d$ are both positive, the inequality

$$\gamma(f_u + g_v) > k^2(d+1).$$

can never be satisfied as it requires that a negative quantity must be greater than a positive quantity. Therefore we have ruled out the possibility that the trace is greater than zero. We instead turn to the determinant and require

$$\det(-Dk^2+\gamma J) = dk^4 - \gamma(df_u+g_v)k^2 + \gamma^2(f_u g_v - f_v g_u) < 0 \tag{7}$$

which will still signify an unstable steady state. We can rewrite the determinant as

$$dK^2 - \gamma(df_u + g_v)K + \gamma^2(f_u g_v - f_v g_u) \tag{8}$$

where $K = k^2$. As $d > 0$, the parabola described by Eq. (8) is concave up. We need the minimum value of Eq. (8) to be negative for the inequality to hold for at least some $K$.

We can immediately spot a weak condition in the inequality. Since in section 3.1 we derived Condition (**C.2**), namely, $(f_u g_v - f_v g_u) > 0$, the only term which can cause the minimum to be negative is the term $-\gamma(df_u + g_v)$. This will only hold under the circumstance that

$$df_u + g_v > 0. \tag{**C.3**}$$

This condition is necessary but not sufficient for the minimum to be below zero. Despite being a fairly weak condition, it still has very interesting implications in terms of the system. Since we enforced $f_u + g_v < 0$ in Condition (**C.2**), we can now write

$$df_u - f_u > df_u + g_v > 0,$$

implying $d > 1$. Since $d$ is the non-dimensional ratio $\frac{D_B}{D_A}$, we see here that an additional condition for Turing pattern generation is that the diffusion coefficients $D_A$ and $D_B$ must be unequal and $D_B > D_A$. Physically, this means that the inhibitor substance must diffuse at a faster rate than the activator substance in order for patterns to form.

As Condition (**C.3**) is not sufficient to guarantee that the minimum is below zero, we proceed by differentiating Eq. (8) and setting equal to 0. This results in

$$2dK - \gamma(df_u + g_v) = 0,$$

meaning the minimum is at $K = \frac{\gamma}{2d}(df_u + g_v)$. Plugging back into Eq. (8), we arrive at

$$K_{min} = \frac{\gamma^2}{4d}\left[4d(f_u g_v - f_v g_u) - (df_u + g_v)^2\right]$$

which we require to be less than 0. Since $d > 0$, this gives us the stronger condition

$$(df_u + g_v)^2 - 4d(f_u g_v - f_v g_u) > 0. \tag{**C.4**}$$

We have now derived all the conditions for Turing instability in regards to unspecified reaction functions $f$ and $g$. In particular, we have two stability conditions,

$$f_u + g_v < 0, \tag{**C.1**}$$

$$f_u g_v - f_v g_u > 0, \tag{**C.2**}$$

and two instability conditions,

$$df_u + g_v > 0. \tag{**C.3**}$$

$$(df_u + g_v)^2 - 4d(f_u g_v - f_v g_u) > 0, \tag{**C.4**}$$

along with the requirements that only one substance is autocatalytic and the inhibiting substance diffuses faster than the activating substance ($d = \frac{D_B}{D_A} > 1$).

## 4. Turing Space

The kinetics derived in Section 3 can be used to inform choices regarding the functions which describe the reaction kinetics so that the system will be capable of generating patterns. In many cases, this means carefully choosing the parameters of the equations. The region in the parameter space in which Turing patterns are capable of forming is known as the Turing space.

### 4.1. Schnakenberg Parameters

Recall the non-dimensional form of the Schnakenberg system given by Eq. (3),

$$\begin{aligned} u_t &= f(u,v) + \nabla^2 u, \\ v_t &= g(u,v) + d\nabla^2 u, \end{aligned}$$

where

$$\begin{aligned} f(u,v) &= \gamma(a - u + u^2 v), \\ g(u,v) &= \gamma(b - u^2 v). \end{aligned}$$

The Turing space here refers to the parameter space of $a$, $b$, and $d$. We can now utilize the conditions for diffusion-driven instability derived in Section 3 to visualize the Turing space for the Schnakenberg kinetics and use that result to inform parameter choices so that the model will be capable of generating patterns.

Again, we start by linearizing about the steady state $(u_*, v_*)$ which is the result of solving

$$\begin{aligned} a - u_* + u_*^2 v_* &= 0 \\ b - u_*^2 v_* &= 0 \end{aligned}$$

and is given by

$$(u_*, v_*) = \left(a + b, \frac{b}{(a+b)^2}\right). \tag{9}$$

The derived conditions depend on the four partial derivatives of functions $f$ and $g$,

$$\begin{aligned} f_u &= -1 + 2uv, \quad & f_v &= u^2, \\ g_u &= -2uv, & g_v &= -u^2. \end{aligned}$$

By substituting these derivatives into the conditions derived in Section 3, we can identify the desired regions in parameter space. In order to plot the conditions in $a$-$b$ space, we parameterize using the steady state given by Eq. (9) by setting

$$\begin{aligned} u_* &= a + b, \\ v_* &= \frac{b}{(a+b)^2} = \frac{u_* - a}{(a+b)^2}. \end{aligned}$$

Now we must go through each of the four conditions and individually asses how they can be met.

### 4.2. Stability Conditions

Condition (**C.1**) states that $f_u + g_v < 0$. Substituting in the partial derivatives for the Schnakenberg kinetics gives

$$-1 + 2uv - u^2 < 0.$$

After parameterizing and solving for $a$, we get a curve defined by

$$a > \frac{1}{2}\left(u_* - u_*^3\right). \tag{10}$$

By the same process, we find that Condition (**C.2**) simplifies to

$$u_*^2 > 0. \tag{11}$$

Clearly, this condition is always met, so long as the concentration of substance $u$ at the steady state is non-zero. Therefore we do not need to consider it going forward.

### 4.3. Instability Conditions

Next, substituting into Condition (**C.3**) results in the requirement that

$$a < \frac{1}{2}\left(u_* - \frac{u_*^3}{d}\right). \tag{12}$$

Since we found that $d > 1$, this condition is compatible with Condition (10) as it will always lie above that curve.

Finally, we consider Condition (**C.4**). Upon substituting in the partial derivatives, parameterizing, and simplifying, we end up with

$$\begin{aligned} a^2\left(4d^2\right) + a\left(4du_*^3 - 4d^2u_*\right) + \\ \left(u_*^6 - 6du_*^4 + d^2u_*^2\right) > 0. \end{aligned} \tag{13}$$

This is a quadratic equation in $a$. The roots are given by

$$a = \frac{-u_*^3 + du_* \pm 2u_*^2\sqrt{d}}{2d}.$$

In order to meet the condition, we need to find the values of $a$ for which the the parabola is positive. The parabola is concave up, so it satisfies the parameterized condition given by Eq. (13), when

$$a < \frac{1}{2}\left(u_* - \frac{u_*^3}{d}\right) - \frac{u_*^2}{\sqrt{d}}, \tag{14}$$

or

$$a > \frac{1}{2}\left(u_* - \frac{u_*^3}{d}\right) + \frac{u_*^2}{\sqrt{d}}. \tag{15}$$

As $\frac{u_*^2}{\sqrt{d}} > 0$, the curve described by Eq. (15) lies above the curve described by Eq. (12). Therefore, it is never possible for both conditions to be met. On the other hand, we see that the curve described by Eq. (14) lies below the curve described by Eq. (12). Therefore we proceed by taking Eq. (14) as stricter upper bound on $a$.

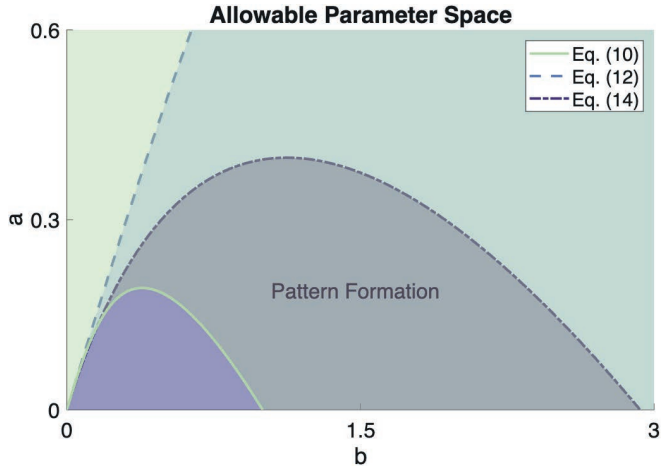



Figure 2: Turing space for $d = 50$. Region where all conditions are met making pattern formation is possible is shown by the darkest shading. Eq. (15) is not pictured as the inequality cannot be satisfied. Condition (11) is also not pictured as it is trivial.

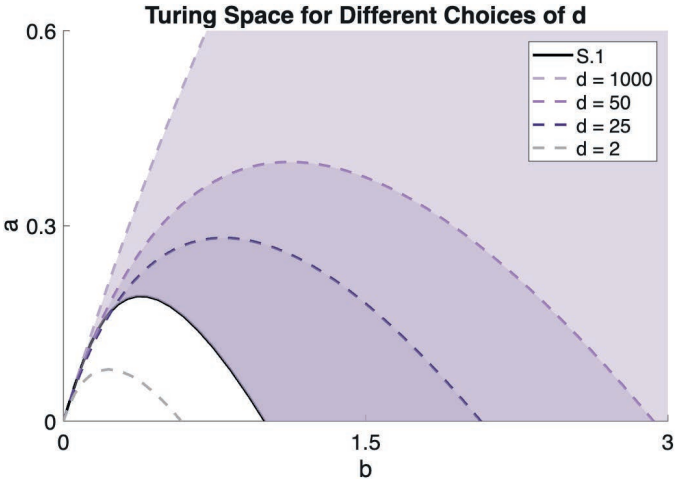



Figure 3: Turing Space for $a$ and $b$ corresponding to various choices of $d$. For each $d$, shading stretches between the dashed curve representing Eq. (S.2) and the solid curve representing Eq. (S.1). Shaded regions represent allowable parameters for different values of $d$. When $d = 2$ as shown by the bottom curve, there is no acceptable region.

We have now gone through all four conditions. To simplify, we found that the Turing space for the Schankenberg model is determined by only two of the four conditions: Eq. (10) and Eq. (14). Denoting the two relevant conditions are denoted with an S for Schnakenberg, we have

$$a > \frac{1}{2}\left(u_* - u_*^3\right), \tag{S.1}$$

which guarantees linear stability in the absence of diffusion and

$$a > \frac{1}{2}\left(u_* - u_*^3\right), \tag{S.2}$$

which guarantees instability in the presence of diffusion.

An example of the relationships between the three non-trivial conditions is plotted in Figure 2. The instability condition, Condition (10), does not depend on choice of $d$ and forms the solid curve. In order for patterns to be formed, $a$ must lie above this curve. The upper curves defined by the instability conditions depend on choice of $d$. Patterns can be formed when $a$ and $b$ lie in between the upper and lower curves. The Turing Space is plotted for different choices of $d$ in Figure 3. If $d$ is too low, there will be no choices of $a$ and $b$ which will be capable of generating patterns. As $d$ grows, the pattern-forming region grows as well.

In conclusion, the allowable parameters fall between the curves defined by Stability Condition (S.1) and Instability Condition (S.2). If we choose parameters in these regions, our model will be capable of generating patterns.

## 5. Pattern Generation

Since the system of partial differential equations defined by the Schankenberg kinetics is non-linear, it does not have a clean analytic solution. Therefore, in order to visualize the patterns, we must turn to numerically approximating the system of equations.

### 5.1. Methods

For simulating the patterns, we use an Implicit Finite Difference scheme. This method is unconditionally stable, second-order accurate in space, and first-order accurate in time. The code used to create all figures in this section can be found in the folder containing the full paper linked below.

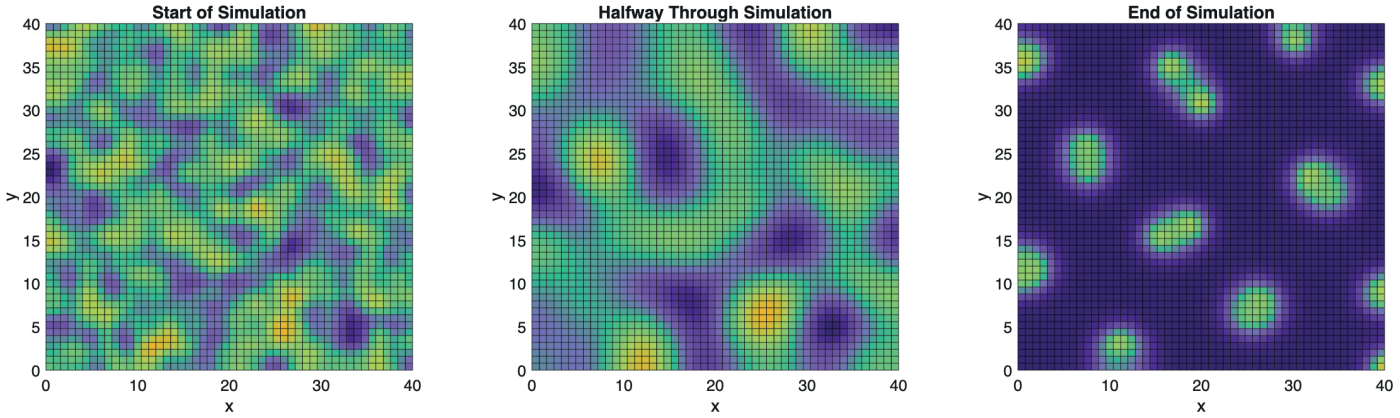
Simulations are implemented on a square defined by $(0, 40) \times (0, 40)$ from time $t = 0$ to time $t = 200$. The number of spatial increments in both the $x$ and $y$ direction is 50 and the number of time increments is 500. We enforce zero-flux boundary conditions given by

$$u_x(0, y, t) = u_x(40, y, t) = v_x(0, y, t) = v_x(40, y, t) = 0,$$
$$u_y(x, 0, t) = u_y(x, 40, t) = v_y(x, 0, t) = v_y(x, 40, t) = 0,$$

The initial condition is given by

$$\begin{aligned} u(x, y, 1) &= u_* + \text{rand}(0, 1) \times 10^{10^{-6}}, \\ v(x, y, 1) &= v_* + \text{rand}(0, 1) \times 10^{10^{-6}}, \end{aligned} \tag{16}$$

where $\text{rand}(0, 1)$ represents a random number in the interval $(0, 1)$ and $u_*$ and $v_*$ are the steady states as in Eq. (9).




(a) Time $t = 20$ (nondim.) (b) Time $t = 250$ (nondim.) (c) Time $t = 500$ (nondim.)

Figure 4: Concentration of substance $u$ over time with random initial condition given by Eq. (16). Parameters are $d = 250$, $a = 0.2$, $b = 1.25$.

The importance of initial condition selection is discussed further in Section 6.3. Simulations are run with the same random seed to guarantee that differences in behavior are results parameter changes and not due to differences in initial condition.

Note also, that the parameter $\gamma$ has been present throughout the derivations of the conditions but not discussed other than noting that it is a positive constant. $\gamma$ is in fact an important parameter in determining the stability of a pattern. Such analysis is briefly touched upon in Section 6.1 but is in large part outside the scope of this paper. For simulations conducted in this paper, $\gamma$ is arbitrarily chosen to be 0.2.

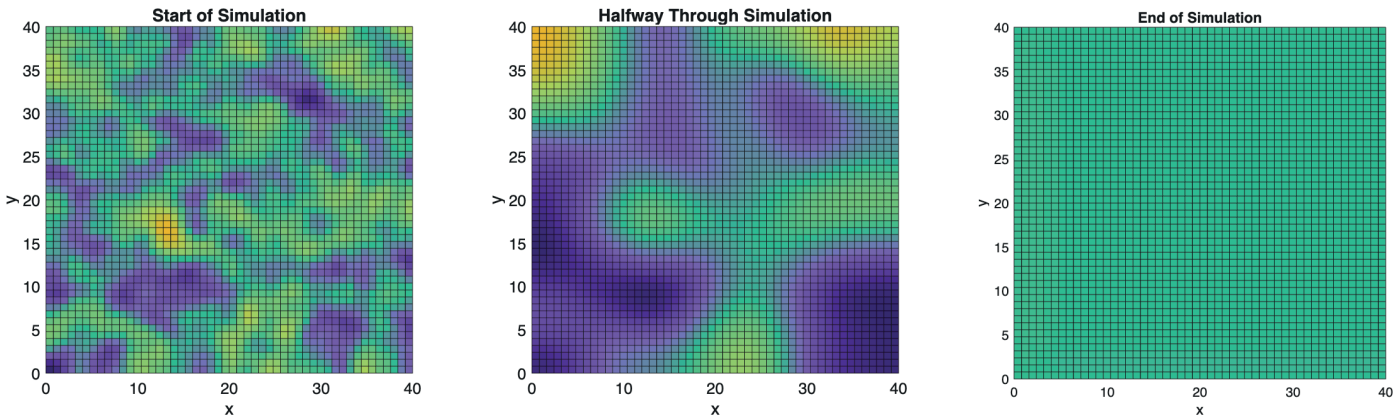
All simulations are conducted on the non-dimensional form of the equations for simplicity, but returning to the dimensional form is a matter of rescaling by the characteristic length introduced in Section 2.3.

### 5.2. Results

We start by selecting parameters which lie within the Turing space as derived in Section 3. Firstly, this means choosing a suitably large value of $d$, the diffusion ratio. $a$ and $b$ must also be carefully selected such that they fall within the region formed by the parameterized conditions relative to the choice of $d$.

A progression over time resulting in a spot pattern formed by substance $u$ is shown in Figure 4. We see that over time, the substance diffuses from the random initial condition to a more amorphous shape around halfway through the simulation. Finally, as time continues to progress, a more defined spot pattern arises.

Substance $v$ also forms a pattern over time which is 180° out of phase with substance $u$ [5]. This is because the patterns as seen from the top are reflective of variations in the concentrations which are actually three dimensional.




(a) Time $t = 20$ (nondim.) (b) Time $t = 250$ (nondim.) (c) Time $t = 500$ (nondim.)

Figure 5: Concentration of substance $u$ over time with random initial condition given by Eq. (16). Parameters are $d = 0.5$, $a = 0.2$, $b = 1.25$.

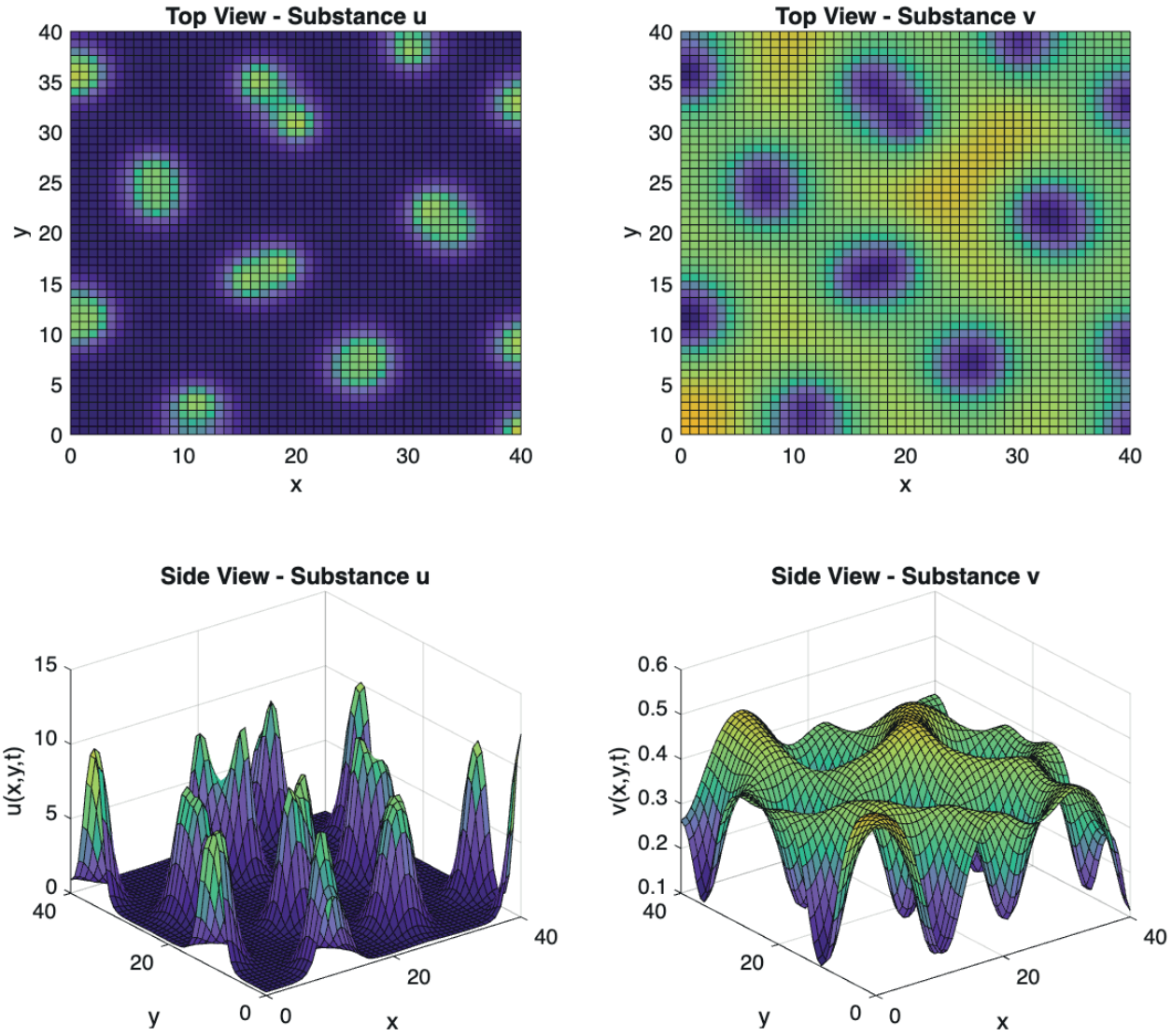



Figure 6: Comparison of substance $u$ and substance $v$ at the end of simulation $t = 500$ (nondim.) Initial condition given by Eq. (16). Parameters are $d = 250$, $a = 0.2$, $b = 1.25$.

So when the concentration of substance $u$ increases in one location, the concentration of substance $v$ decreases there. This is shown both from the top and side in Figure 6.

Again calling back to the linear stability analysis, if the parameters do not lie in the Turing space, the system will not be capable of generating patterns. An example is shown in Figure 5 where the activator substance $u$ diffuses faster than the inhibitor substance $v$, leading to diffusion ratio $d$ which is less than 1. The simulation begins with the same initial condition, but this time, we see that the amorphous halfway state ends up stabilizing to a homogeneous state by the end of the time period. This is exactly what we expect to see. When the system does not meet the conditions for Turing instability, diffusion plays its typical role and stabilizes or "flattens" the concentrations of each substance.

## 6. Discussion

In this paper, we conducted a linear stability analysis to derive the necessary conditions for Turing instability for reaction-diffusion systems of the form of Eq. (2). This analysis provided us not only with mathematical conditions, but also with physical insights into what types of systems are capable of producing spatial patterns. We found that only one of the two substances can be autocatalytic for patterns to form. We also find that the substance which is not autocatalytic (the inhibitor) must diffuse at a faster rate then the other substance for the system to experience Turing instability.

We then applied the derived conditions to the Schnakenberg kinetics in order to define the Turing space and choose parameters which are capable of generating patterns. Us-

ing numerical simulations, we observed pattern formation in the cases where parameters met the conditions and the expected progression to the homogeneous steady state in the cases where parameters did not meet the conditions.

### 6.1. Pattern Stability

Until this point, we have analyzed the linearized version of the model. While this allows us to make informed parameter choices and predict some information about the formation of patterns, adding the nonlinearities back into the system leads to less predictable results. Though we are able to generate patterns based on the linear stability analysis, one thing our results have not provided us with is the ability to say definitively whether or not the patterns will become stable over time. This can be deduced with further analysis. Then parameters can be selected such that the wavenumber as defined by the linear analysis grows so rapidly that it balances out the contribution from the nonlinear terms. Texts such as *Mathematical Biology*[2] conduct this analysis in order to more accurately predict which parameter values will cause the formations of stable patterns.

Of course, stable pattern formation is a key point of interest for many biological models, since patterns in nature such as vegetation growth and animal spots often appear fairly stagnant. Other approaches to modeling realistic time dynamics are also the topic of many works in Turing pattern modeling. For example, in their paper *Two-stage Turing model for generating pigment patterns on the leopard and the jaguar*, Liu et al. investigate the dynamics of leopard spot patterns evolving as the leopard ages and implement with more biologically realistic time dynamics for pattern formation [10].

### 6.2. Selection of Reaction Kinetics

In this paper we analyze the Schnakenberg reaction kinetics which are known to favor spot patterns. When developing a Turing pattern model for some natural phenomenon, it is important to define the reaction kinetics based on the desired type of pattern. Generally, there are two major pattern groups: spots and stripes. These are reflected in the equations representing the reaction kinetics. A cubic term is known to generate stripe patterns but the presence of a quadratic term will dominate and favor spots [8]. It has been proven that though spot and stripe patterns may emerge in the same space as the dynamics of the reaction-diffusion system play out over time, these two pattern types are mutually exclusive as stable patterns [8]. Going back to the chemical and biological undertones to these reaction-diffusion equations, we note that it is rare to see an animal with a pattern consisting of intermixed spots and stripes.

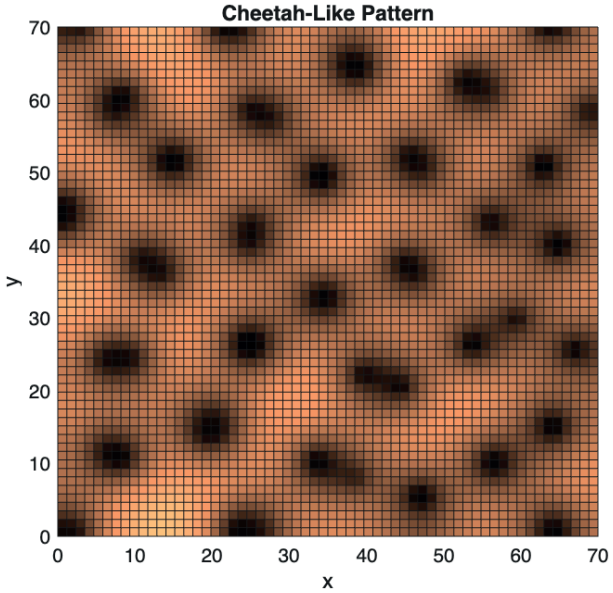



Figure 7: Pattern resembeling cheetah coat generated by Schnakenberg Kinetics. Initial condition given by Eq. (16). Parameters are $d = 250$, $a = 0.2$, $b = 1.25$, $\gamma = 0.2$. Simulation run on a square domain defined by $(0, 70) \times (0, 70)$ (nondim.) with 60 spatial increments. Time spans from $0 < t < 500$ (nondim.) with 200 time increments.

### 6.3. Initial and Boundary Conditions

Turing patterns are very sensitive to initial condition in the sense that small changes in initial condition can cause significant changes in the frequency and size of the pattern [5]. The initial condition generally does not impact the ability of the system to produce patterns and the type of patterns that will be produced. That being said, the linear analysis performed in the previous sections assumes the system to be locally linear, meaning that if the initial condition is too far from the homogeneous steady state Eq. (9), then there is no guarantee that we will see Turing instability [5].

The size of the domain can have a similarly large effect. Although the type of pattern does not change, using too a small domain may make a spot pattern appear as stripes or look as if no pattern formed when it really formed outside of the domain [2][5]. Of course, patterns in nature are usually found on significantly more complex domains than the squares we have been considering in this paper. In fact, examining different domains and boundary conditions for those domains is a widely studied extension of the Turing pattern model.

### 6.4. Broader Applications of Turing Patterns

One common application of Turing patterns is in modeling animal patterns. This is an especially exciting appli-

cation, as it has been shown that morphogenesis of animal cells is a likely cause of these patterns [3][4]. In their paper *Diffusiophoresis-enhanced Turing patterns*, Alessio et al. study pattern formation in different species of fish and argue that the similarities arising from reaction-diffusion equations suggest that they are the dominant biological process at play [4].

Since the Schnakenberg kinetics form spot patterns, they are suitable for modeling the coats of animals such as cheetahs. Applying the analysis performed in Section 3 to selecting parameters which will generate patterns results in the Cheetah-like pattern shown in Figure 7.

Many other animal patterns have been modeled using Turing patterns as well. For example, with a more complex model, Liu et al. successfully model the rosette patterns found on leopards and jaguars [10]. Other patterns such as those found on seashells, zebras, and giraffes are also common applications of Turing pattern models [2].

Of course, it is difficult to validate such models, since the spot patterns of every animal are different and will not be exactly the same as any generated pattern. In order to draw conclusions about the biological plausibility of their model, studies such as the one by Alessio et al. [4] and that of Liu et al. [10] conduct specific measurements on various aspects of the biological and simulated patterns. These measurements often involve looking at pattern widths, lengths, sizes, and frequencies.

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## LIZA PLATONOV



As a kid, I had a fairly complicated relationship with math. My dad is a high school math teacher, and he was very committed to making sure that I had a strong foundation in math. For most of elementary and middle school, my dad and I would do between 30 minutes and two hours of supplementary math every day at home. He also had some "unconventional teaching methods," typically pushups and situps. When I reached high school, I finally got some reprieve from the at-home math lessons, and I started to adopt a more casual attitude toward the subject. In college, I immediately committed to a computer science major, but math slowly started to creep back in, and junior year I ended up adding a math major.

I'm so glad that I found the math department in the end. It's been really rewarding to me to be able to use the tools that I've been building up since elementary school in much more complex applications. One of my favorite things about math is the way that everything is connected. Every time I take a new class or learn about a new concept I can find ways that it relates to something else I have learned. I may have joined the math major relatively late in my Middlebury career, but I've learned so much and I'm very grateful for the community I have in the math department. I've made so many great friends through the math department and the way that the math students here always help each other out has really contributed to my love for my math major. Next year, I'm starting a math PhD at Univeristy of Utah, and I'm forever grateful to the Middlebury math department for helping me discover my interest in math.